



Final Round 2026

First Exam – Solutions

Duration: 4 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Aarburg

March 13, 2026

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

1. Let n be a positive integer. Archi and Baldibus are constructing a number on a blackboard, one digit at a time. To start, Archi writes down one of the **nine** digits from 1 to 9. Then, starting with Baldibus, the two players alternate the following:

- Baldibus appends one of the **ten** digits from 0 to 9 to the **end** of the current number.
- Archi prepends one of the **nine** digits from 1 to 9 to the **front** of the current number.

Archi wins if, immediately after Archi writes a digit, the number on the board is divisible by n . Determine the values of n for which Archi can guarantee a win, no matter what Baldibus does.

Solution

Answer: Archi wins for $1 \leq n \leq 9$, and loses for $n \geq 10$.

Proof: We call Archi A and Baldibus B . A easily wins for all $n \leq 9$ by just writing n as his first number and winning instantly.

If A is now to win for $n \geq 10$, he will need to win immediately after one of B 's moves, since A cannot win on the first turn anymore. A then needs to fulfill:

$$n \mid a10^{2k+1} + 10X + b, \quad (1)$$

where a is A 's last digit written, b is B 's last digit written, and X is the number written on the board before B 's last move (and k is an arbitrary positive integer). So all in all, if A wants to win, there needs to be an integer X on the board, such that for any digit b that B can write on the board, A can write a digit $a(b)$ such that (1) holds. Notice now that B has 10 choices for b and A only has 9 for a . We now show that:

For two different choices of b , the response $a(b)$ from A needs to be different as well. Indeed, let for the sake of contradiction $n \mid a(b_1)10^{2k+1} + 10X + b_1$ and $n \mid a(b_2)10^{2k+1} + 10X + b_2$ for two different $b_1 \neq b_2$ with $a(b_1) = a(b_2) = a$. Then we would also have

$$n \mid (a10^{2k+1} + 10X + b_1) - (a10^{2k+1} + 10X + b_2) = b_1 - b_2.$$

But since n is bigger or equal to 10 and divides $b_1 - b_2$, which in absolute value is strictly smaller than 10, this is only possible with $b_1 = b_2$, which is a contradiction. This means that to win for $n \geq 10$, A needs to have 10 different responses to B 's 10 possible different choices of b , which is impossible.

Marking Scheme

- (A) **1P:** Claiming the solution and proving the cases $1 \leq n \leq 9$.
- (B) **1P:** Focusing on the 2 last moves of A and B before the end of the game.
- (C) **5P:** Claiming and proving that A cannot have a countermove for each of B's moves. (for $n \geq 10$)
 - (C1) **1P:** Working with some sort of countermove for A that depends on B's last move.
 - (C2) **4P:** Proving that A cannot have a countermove for each of B's moves.

If at most one point has been claimed, another point can be obtained by showing some specific $n \geq 10$ does not work.

-2P: For lack of justification when it comes to explaining why Baldibus's choice lead to distinct situations.

-1P: For misunderstanding the small cases of n .

2. Let $n, k \geq 2$ and $a_1, a_2, \dots, a_k$ be positive integers satisfying $n < k < 2n$ and

$$2^n - 1 \mid 2^{a_1} + 2^{a_2} + \dots + 2^{a_k}.$$

Show that at least three of $a_1, a_2, \dots, a_k$ have the same remainder when divided by n .

Solution

Let $M = 2^n - 1$. Clearly the order of 2 mod M is n , hence we may reduce all the a_i modulo n . Counting the powers of 2 by their amount of appearances, we now arrive at the following equivalent problem:

Let $b_0, b_1, \dots, b_{n-1} \in \{0, 1, 2\}$ such that $n < b_0 + \dots + b_{n-1} < 2n$ and $2^n - 1 = b_0 + 2b_1 + \dots + 2^{n-1}b_{n-1}$, where we may assume equality since if $b_0 + 2b_1 + \dots + 2^{n-1}b_{n-1} \geq 2(2^n - 1)$ then at least one of the b_i would need to exceed 2 (if they were all 2, their sum would be equal to $2n$, contradicting the condition $b_0 + \dots + b_{n-1} < 2n$).

Let $c_i = b_i - 1$ for all i , thus $c_i \in \{-1, 0, 1\}$, we can again reformulate the problem to $0 = c_0 + 2c_1 + \dots + 2^{n-1}c_{n-1}$ by subtracting $1 + 2 + \dots + 2^{n-1}$ from both sides of the equation. We have $0 < c_0 + \dots + c_{n-1} < n$ since $1 + 2 + \dots + 2^{n-1} = 2^n - 1$.

To see why this cannot be, consider the highest indexed c_j not equal to 0, then notice that $c_0 + c_1 2 + c_2 2^2 + \dots + c_{j-1} 2^{j-1} = -c_j 2^j$, which is impossible since $|c_i| \leq 1$ and $1 + 2 + \dots + 2^{j-1} = 2^j - 1$, hence $c_i = 0$ for all i , contradicting $0 < c_0 + \dots + c_{n-1}$.

Alternative finish from the same setup:

We "merge together" powers of 2 that appear multiple times. Formally, if $b_i > 2$, where i is maximal then we replace (b_i, b_{i+1}) by $(b_i - 2, b_{i+1} + 1)$. Since the sum of the b_i is strictly decreasing, this process must eventually terminate. Furthermore, no coefficient outside of $b_0, \dots, b_{n-1}$ will be touched (so b_n will not suddenly increase) since the sum is invariant.

At the very end of the process, $b_0 = b_1 = \dots = b_{n-1} = 1$. There are now multiple ways to see why this cannot be:

- At the very start, look at the lowest index which is not 1. Since every step only operates on an index and ones higher than it, this index will perform at most one operation, ergo if it is 0 at the start, it will remain 0, and if it is 2 at the start, it will turn to 0 and stay at 0, so we won't be able to reach the desired goal value
- Note that after the very first step, at least one of the $b_i = 0$ (the one we just merged). Now we claim that if there is a 0 on a given step, there will be a 0 on the step afterwards. Indeed, if we currently perform an operation on a index of size > 2 , then we previously must have performed an operation on a lower index, whilst this one was at 2, contradicting us always picking the largest index. If the current index is 2, then it will be 0 after the operation.

By this, there will always be an index of size 0, unless no operations are performed, contradicting $n < b_0 + \dots + b_{n-1}$ by PHP

Marking Scheme

In total, the sum of points obtained in Part A and Part B is awarded. The parts are NOT additive within themselves except A1 and A2, which are not additive with A3 though.

(A) **3P:** Part A - Reformulating the Problem:

(A1) **1P:** Noting that $2^n \equiv 1 \pmod{2^n - 1}$, and remarking that this let's us change the a_i by n without changing the expression modulo $2^n - 1$

(A2) **1P:** Noting that $\sum 2^{a_i}$ for a_i reduced modulo n must equal $2^n - 1$ and is not just divisible by it

(A3) **3P:** Reformulating the Problem in a useful way, which sets up for continuation, e.g. counting the powers of 2 by multiplicity.

(B) **4P:** Part B - Finishing:

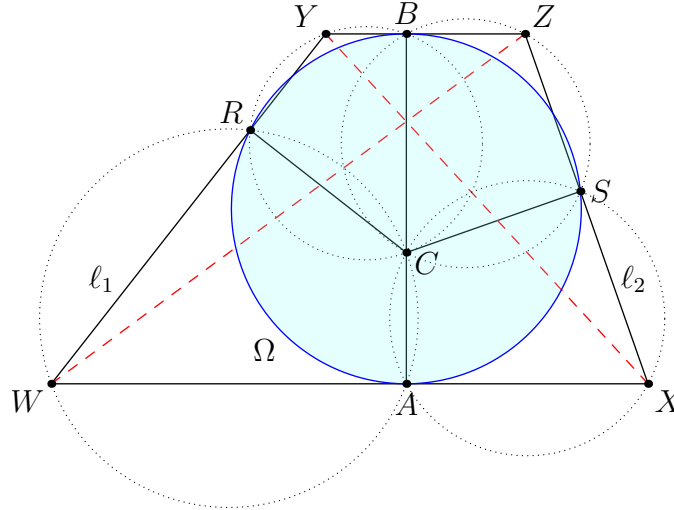
(B1) **2P:** Considering the biggest coefficient that is not 1 in the power-of-2 expansion.

(B2) **2P:** Combining equal powers of 2 and showing this process terminates.

(B3) **4P:** Finishing.

3. Let Ω be a circle with diameter $[AB]$ and C a point on the segment $[AB]$. Let ℓ_1 and ℓ_2 be two lines, such that the projections of C onto them both lie on Ω . Let W and X be intersections of the tangent to Ω at A with ℓ_1 and ℓ_2 respectively. Let Y and Z be intersections of the tangent to Ω at B with ℓ_1 and ℓ_2 respectively. Prove that WZ , XY and AB concur at a point.

Solution



Note that, as AB is a diameter of Ω and WX , YZ are tangents, we have $WX \perp AB \perp YZ$. Therefore, if P is the intersection of WZ and AB , the right triangles PAW and PBZ are similar (they share all angles), and therefore, $\frac{AP}{PB} = \frac{AW}{ZB}$ (their sides are proportional). Similarly, if Q is the intersection of XY and AB , we have $\frac{AQ}{QB} = \frac{AX}{YB}$. We wish to prove that $P = Q$, that is, $\frac{AW}{ZB} = \frac{AX}{YB}$, or equivalently $\frac{AW}{AX} = \frac{BZ}{BY}$ (which translates into the existence of a homothety with center $P = Q$, sending A to B , W to Z and X to Y). After rearranging, this boils down to proving that

$$AW \cdot BY = AX \cdot BZ.$$

Let R be the foot of C on ℓ_1 and S be the foot of C on ℓ_2 . Due to the right angles, the quadrilaterals $CAWR$, $CAXS$, $CBYR$ and $CBZS$ are all cyclic. This allows us to compute that

$$\angle YCW = \angle YCR + \angle RCW = \angle YBR + \angle RAW = \angle BAR + \angle RBA = 90^\circ,$$

where we respectively use the tangent chord lemma with BY and AW , and the fact that ARB is a right triangle at R because AB is a diameter and $R \in \Omega$. Similarly, $\angle XCZ = 90^\circ$, meaning that YCW and XCZ are right triangles at C .

But then, the right triangles CAW and YBC are similar: indeed, they have equal angles since

$$\angle AWC = 90^\circ - \angle WCA = 180^\circ - \angle YCW - \angle WCA = \angle BCY.$$

Therefore, the ratios of their sides are equal and we get

$$\frac{CA}{AW} = \frac{YB}{BC} \implies AW \cdot BY = AC \cdot CB.$$

By the same reasoning on ℓ_2 , triangles CAX and ZBC are similar, and we get that

$$\frac{CA}{AX} = \frac{ZB}{BC} \implies AX \cdot BZ = AC \cdot CB.$$

Putting everything together, we get the desired equality of products, which allows us to conclude.

Marking Scheme

A student gets awarded the sum of their points for parts (A) and (B). These parts are themselves non-additive.

(A) +**5P**: Proving that $AW \cdot BY = AX \cdot BZ$

(A1) **1P**: Noting that at least one of $CAWR, CAXS, CBYR, CBZS$ is cyclic

(A2) **2P**: Proving that $\angle YCW = 90^\circ$ (and/or $\angle XCZ = 90^\circ$)

(A3) **2P**: Deriving the equality $\angle AWC = \angle BCY$ (or a symmetric equality)

(A4) **3P**: Proving that CAW and YBC are similar (and/or CAX and ZBC)

(A5) **4P**: Proving only $AW \cdot BY = AC \cdot CB$ (or $AX \cdot BZ = AC \cdot CB$)

(B) +**2P**: Concluding (rephrasing concurrency)

(B1) **1P**: Proving that $\frac{AP}{PB} = \frac{AW}{ZB}$

(B2) **1P**: Claiming that $\frac{AW}{AX} = \frac{BZ}{BY}$

(B3) **1P**: Justifying why concurrency can be rephrased in terms of the existence of a homothety

(B4) **2P**: Justifying why concurrency can be rephrased in terms of an equality of ratios

To have a complete solution, the students are **not** asked to justify why the concurrency point lies on or outside of the segment $[AB]$ depending on the relative position of ℓ_1 and ℓ_2 (only treating one of these two cases suffices).

4. Determine all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that for all $x, y \in \mathbb{R}$,

$$f(x+y) + f(f(y)x+y) = f(x+2y) + f(y)x.$$

Solution

Answers: $f \equiv 0$ and $f(x) = x$ (we can easily check that both are valid solutions).

As usual, denote by $P(x, y)$ the initial equation. Plugging in $x = 0$ gives us $f(2y) = 2f(y)$ $(\star)$. In particular $f(0) = 0$. Now $P(x, y/2)$ gives us

$$f(x+y/2) + 1/2f(f(y)x+y) = f(x+y) + 1/2f(y)x,$$

where we got the factors $1/2$ out of the parentheses thanks to the previous observation $(\star)$. But we can subtract $1/2P(x, y)$ from the above expression to get

$$f(x+y/2) - 1/2f(x+y) = f(x+y) - 1/2f(x+2y) \quad (\text{after cancelling terms out}).$$

Multiplying the entire equation by 2 and using $(\star)$, we get

$$f(2x+y) + f(x+2y) = 3f(x+y).$$

But for any reals s, t , we can find x, y such that $2x+y = s$ and $x+2y = t$. That means that $f(s) + f(t) = 3f(\frac{s+t}{3})$. And, by further substituting $t = 0$, we get $3f(\frac{s}{3}) = f(s)$, for all $s \in \mathbb{R}$. We use this observation on the previous equality, to get

$$f(s) + f(t) = f(s+t) \quad \forall s, t \in \mathbb{R}$$

(meaning that f is additive).

And from there we are essentially done; looking at the original expression and using additivity we eventually get $f(f(y)x) = f(y)x$, so there are two cases:

- either f is identically zero (which is a valid solution),
- or there is a u such that $f(u) \neq 0$, and hence $f(u)x$ ranges over all the reals, meaning that $f(x') = x'$ for all $x' \in \mathbb{R}$.

Second solution (Francesco):

Suppose that our function isn't the constant zero function (which is clearly a solution).

- by $P(0, 0)$: $2f(0) = f(0) \implies f(0) = 0$
- by $P(0, y)$: $2f(y) = f(2y)$ for all real numbers y

Moreover, by $P(2x, 2y)$ we get

$$f(2x+2y) + f(2xf(2y)+2y) = f(2x+4y) + 2xf(2y).$$

Using our previous result, we obtain:

$$2f(x+y) + 2f(2xf(y)+y) = 2f(x+2y) + 4xf(y).$$

and subtracting $2P(x, y)$ yields

$$\begin{aligned} 2f(2xf(y)+y) &= 2f(xf(y)+y) + xf(y) \\ \iff f(2xf(y)+y) &= f(xf(y)+y) + xf(y). \end{aligned}$$

Suppose that $\exists a \in \mathbb{R}^*$, such that $f(a) = 0$. Then by $P(x - a, a)$:

$$f(x) + 0 = 0 + f(a + x) + 0 \implies f \text{ has period } a.$$

Since f isn't the constant zero function $\exists x_0 \in \mathbb{R}$, s.t $f(x_0) \neq 0$.

Then replacing x by $\frac{a}{f(x_0)}$ and y by x_0 in our previous result :

$$f(2a + x_0) = f(a + x_0) + x_0.$$

Since f has period a :

$$f(2a + x_0) = f(a + x_0) \implies x_0 = 0,$$

which is a contradiction since $f(x_0) \neq 0$, thus $f(a) = 0 \iff a = 0$.

So back to our last equation: let $z \neq 0$ and $x = \frac{z}{f(-z)}$, $y = -z$ (since $z \neq 0 : f(-z) \neq 0$) we get : $f(2z - z) = f(z - z) + z \implies f(z) = z \forall z \in \mathbb{R}^*$, and since $f(0) = 0$, we indeed have that $f(x) = x \forall x \in \mathbb{R}$. So our two solutions are the only valid ones, and we can check that they indeed work.

Marking Scheme

First solution:

- (A) **+1P:** Showing $f(2y) = 2f(y)$
- (B) **2P:** Computing $P(x, \frac{y}{2}) - \frac{1}{2}P(x, y)$
- (C) **3P:** Finding $f(x + 2y) + f(2x + y) = 3f(x + y)$
- (D) **5P:** Showing additivity, i.e. $f(x + y) = f(x) + f(y)$

Only the point A is additive to the other points.

Second solution:

- (A) **+1P:** Showing $f(2y) = 2f(y)$
- (B) **+1P:** finding that f has period a for a a zero f
- (C) **2P:** Computing $P(2x, 2y) - 2P(x, y)$
- (D) **3P:** Concluding that $f(x) = 0 \Leftrightarrow x = 0$
- (E) **4P:** Finding $f(2x + y) = f(x + y) + x$ for all $x \in \mathbb{R}$ and $y \neq 0$.

For partial solutions, only one of the items (C), (D), (E) can be awarded (but items (A) and (B) can be awarded additionally).



Final Round 2026

Second Exam – Solutions

Duration: 4 hours

Difficulty: The problems are ordered by difficulty.

Points: Each problem is worth 7 points.

Aarburg

March 14, 2026

Preliminary Remark

A complete solution is worth 7 points. For every problem, up to 2 points can be deducted from a full solution for (minor) flaws. Partial marks are attributed according to the marking schemes. In the case of multiple marking schemes for the same problem, the score is the maximum among all the marking schemes.

5. We call a strictly increasing sequence $a_1, a_2, \dots, a_{2026}$ of positive integers *indivisible* if for all $2 \leq k \leq 2026$, the sum of any k consecutive terms is **not** divisible by k .

Prove that an indivisible sequence exists and determine the smallest possible value of the last term in an indivisible sequence.

Solution

There exists an indivisible sequence and the smallest possible value of the last term is 4050.

Proof. We claim that the sequence $b_i := 2i - 2 + \frac{(-1)^{i+1}+1}{2}$ is indivisible. Note that for i odd we have $b_i = 2i - 1$ and for i even we have $b_i = 2i - 2$, we can quickly check that $(b_i)_{1 \leq i \leq 2026}$ is strictly increasing. Now for any $k \geq 2$ and $1 \leq i \leq 2027 - k$ we have

$$\sum_{j=i}^{i+k-1} b_j = \sum_{j=i}^{i+k-1} (2j - 2) + c_{i,k} = k^2 + 2ik - 3k + c_{i,k} \equiv c_{i,k} \pmod{k},$$

where $c_{i,k}$ is the amount of odd numbers in the set $\{i, i+1, \dots, i+k-1\}$, so is equal to $\lceil \frac{k}{2} \rceil$ or $\lfloor \frac{k}{2} \rfloor$. As k divides neither $\lceil \frac{k}{2} \rceil$, nor $\lfloor \frac{k}{2} \rfloor$ for all $k \geq 2$, it follows that $(b_i)_{1 \leq i \leq 2026}$ is indeed indivisible, with $b_{2026} = 4050$.

Let $a_1, a_2, \dots, a_{2026}$ be an indivisible sequence. As it is a strictly increasing sequence of positive integers we have $1 \leq a_1$ and $a_i + 1 \leq a_{i+1}$ for all $1 \leq i \leq 2025$.

Claim. For all $1 \leq i \leq 2024$ we have $a_{i+2} \geq a_i + 4$.

Using $a_2 \geq a_1 + 1 \geq 2$, we use the claim to show by induction that $a_{2n} \geq 4n - 2$, so in particular $a_{2026} \geq 4050$.

From the property of an indivisible sequence we have

$$a_i + a_{i+1} \not\equiv 0 \pmod{2} \quad \forall 1 \leq i \leq 2025 \quad (2)$$

$$a_i + a_{i+1} + a_{i+2} \not\equiv 0 \pmod{3} \quad \forall 1 \leq i \leq 2024. \quad (3)$$

It follows that any two consecutive elements a_i, a_{i+1} have different parity, so $a_{i+1} = a_i + 1$ or $a_{i+1} \geq a_i + 3$. If the later is the case we immediately have $a_{i+2} \geq a_{i+1} + 1 \geq a_i + 4$. Similarly if we have $a_{i+2} \geq a_{i+1} + 3$, the claim would follow.

Assume now that there exists an i such that $a_{i+2} < a_i + 4$. From the above observation we must have $a_{i+2} = a_{i+1} + 1$ and $a_{i+1} = a_i + 1$. But then we have

$$a_i + a_{i+1} + a_{i+2} = a_i + (a_i + 1) + (a_i + 2) = 3(a_i + 1) \equiv 0 \pmod{3},$$

a contradiction to the second equation (3). So our claim holds true for all $1 \leq i \leq 2024$ and we have $a_{2026} \leq 4050$.

Marking Scheme

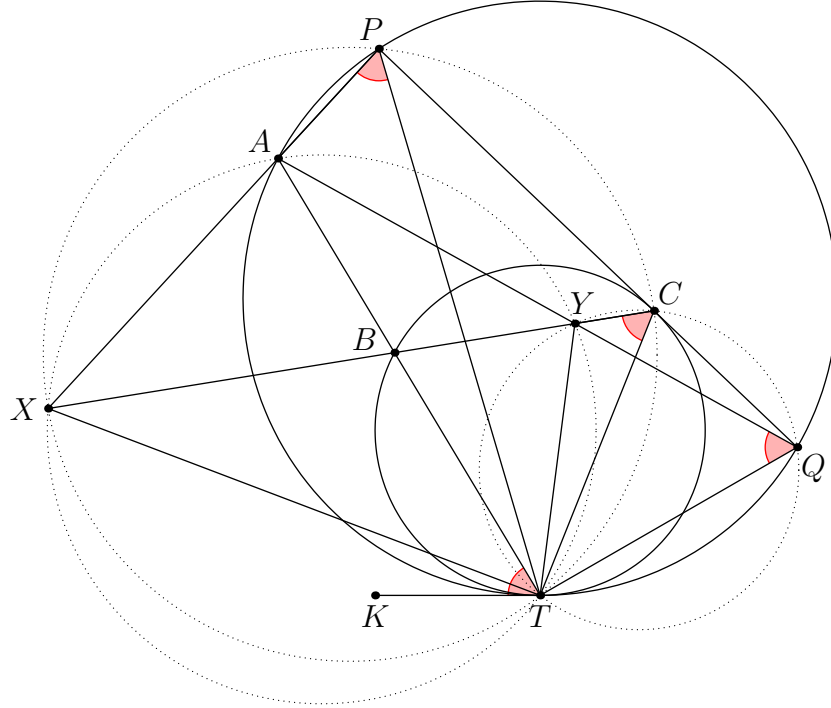
- (A) **1P:** Give an example of an indivisible sequence with $a_{2026} = 4050$.
- (B) **1P:** Prove that the given sequence is indivisible.
- (C) **4P:** Prove that $a_{i+2} \geq a_i + 4$
 - (C1) **1P:** Claim a enumerative bound between non consecutive elements stronger than what follows from monotonicity.
 - (C2) **1P:** Show that $a_{i+1} = a_i + 1$ or $a_{i+1} \geq a_i + 3$.
 - (C3) **1P:** Show that if $a_{i+2} \leq a_i + 3$ then $a_i + 2 = a_{i+1} + 1 = a_{i+2}$.
- (D) **2P:** Prove that $(1, 2, 5, 6, 9, \dots, 4050)$ is the sequence achieved when the elements are sequentially chosen to be the smallest possible (greedy algorithm).

The partial points awarded are calculated as follows:

$$A + B + \max(C, D).$$

6. Let Γ_1 and Γ_2 be two circles tangent at T , such that Γ_2 is inside of Γ_1 . Let PQ be a chord of Γ_1 , tangent to Γ_2 at C . Point $A \neq T$ lies on Γ_1 and AT intersects Γ_2 at $B \neq T$. Let line BC intersect AP and AQ at X and Y , respectively. Prove that TB bisects the angle $\angle XTY$.

Solution



Let K be a point on the common tangent at T . Then, by the tangent chord theorem,

$$\angle YCT = \angle BCT = \angle BTK = \angle ATK = \angle AQT = \angle YQT,$$

so $CQTY$ is cyclic. Similarly,

$$\angle XCT = \angle BCT = \angle BTK = \angle ATK = \angle APT = \angle XPT,$$

so $CPXT$ is cyclic. Using that $CQTY$ is cyclic, we also have that

$$\angle XYT = \angle CQT = \angle PQT = \angle XAT$$

and $AXTY$ is cyclic. Using the three cyclic quadrilaterals, we obtain

$$\angle YTA = \angle YXA = \angle CXP = \angle CTP$$

and

$$\angle ATX = \angle AYX = \angle QYC = \angle QTC,$$

so it suffices to show $\angle QTC = \angle CTP$. This is true by the shooting lemma, as TC intersects arc PQ at its midpoint.

Remark: For the above solution to hold in all configurations, directed angles are needed.

Marking Scheme

- (A) **5P:** Proving that it suffices to show $\angle QTC = \angle CTP$.
 - (A1) **1P:** Proving that one quadrilateral is cyclic among $CQTY, CPXT, AXTY$.
 - (A2) **2P:** Proving that two quadrilaterals are cyclic among $CQTY, CPXT, AXTY$.
 - (A3) **3P:** Proving that all three quadrilaterals $CQTY, CPXT, AXTY$ are cyclic.
- (B) **1P:** Proving that $\angle QTC = \angle CTP$, or equivalently that TC bisects arc PQ .

A complete solution is awarded 7 points. No points are awarded for claims, and no points are deducted for not using directed angles. If no other points are granted, it is worth 2P to prove that $AX = AY$.

7. There are finitely many integers on a board, all of which are at least 2. Pingu repeatedly carries out the following step: first, Pingu chooses an integer $n \geq 2$ that is coprime to every integer currently on the board. Pingu then replaces each integer x on the board with the unique integer $y \in \{1, 2, \dots, n-1\}$ such that $xy \equiv 1 \pmod{n}$.

Pingu stops as soon as there is a 1 anywhere on the board at the end of a step. If at this point, there is also a number different from 1 on the board, Pingu loses. Otherwise, if all the numbers are 1, Pingu wins. Prove that Pingu can always win, regardless of the starting integers.

Example: In a step, if Pingu chooses $n = 11$, Pingu replaces 3, 8, 13, 8 with 4, 7, 6, 7.

Solution

Solution 1

We first want to make all numbers on the board odd and then finish by taking $n = 2$.

Let P be the product of all the numbers on the board. Taking $n = 2P + 1$, we can show that $y = \frac{(x-1)n+1}{x}$ for any x on the board:

First, obviously $xy \equiv 1 \pmod{n}$. Second, note that $\frac{(x-1)n+1}{x} = \frac{(x-1)(2P+1)+1}{x} = \frac{2Px-2P+x}{x} = 2(x-1)\frac{P}{x} + 1$. It's easy to see that this is an odd integer between 2 and $n-1$, so exactly what we wanted.

Solution 2

The idea is the same as in Solution 1. We first make all the integers odd (and greater than 1). Then, we can use $n = 2$ to finish.

We claim Pingu can make any given even integer odd, whilst keeping all the odd integers odd. Let S be the set of numbers on the board. For an integer x , let $\text{odd}(x)$ denote its largest odd divisor (in other words, $\text{odd}(x) = x \cdot 2^{-\nu_2(x)}$). Let $M = \prod_{s \in S} \text{odd}(s)$ (as you will see we could actually take a lowest common multiple instead of a product here, but no need to bother).

Now, suppose not all the integers are odd. Pick an even integer a on the board, and define $b = \text{odd}(a)$, $\ell = \nu_2(a)$ - in other words, $a = 2^\ell b$.

Now, by CRT, pick an integer n (larger than all integers on the board) such that:

- $n \equiv -\frac{1}{2} \pmod{M}$;
- $n \equiv \frac{2^\ell - 1}{b+2} \pmod{2^{\ell+1}}$;
- n is of course coprime with all integers on the board.

The denominators in the fractions above are both coprime with the moduli class in question, so no issues there.

Note that nothing will be sent to 1 because n is larger than all the integers on the board.

By the first condition, we know that $M \mid 2n+1$, and obviously $(2n+1, n+1) = 1$, so any divisor of M will divide $2n+1$ and not $n+1$. It follows for any odd numbers x on the board, its inverse modulo n is $(2n+1)/x$, hence also odd.

What is the inverse of a ? We claim that

- $a \mid (b+2)n+1$,
- $\nu_2((b+2)n+1) = \ell$

Together, these two facts would imply that the inverse of a is odd for $a > 2$. To see this, note that the inverse of a is $((b+2)n+1)/a$ (it does not divide any earlier number of the form $nk+1$ because $(a,n)=1$ and $b+2 < a$) and since $\nu_2((b+2)n+1) = \nu_2(a)$, this quotient is odd.

Now if $a = 2$ we can still use the same n once (with an additional condition $n > 3$ and all the odd numbers stay odd and a gets transformed into a number bigger than 2. If it becomes odd we are done, else we repeat the step and it becomes odd.

Now we prove the facts. Note that since $b \mid M \mid 2n+1$, it's easy to see that $b \mid (b+2)n+1$. So all we need is that $\nu_2((b+2)n+1) = \ell$, since this is already what the second fact says and also would show that $2^\ell \mid (b+2)n+1$, finishing the first fact. In terms of congruences, this is equivalent to $(b+2)n+1 \equiv 2^\ell \pmod{2^{\ell+1}}$, which after rearranging is just the above.

Marking Scheme

A student gets awarded the sum of their points for parts (A), (B) and (C).

- (A) **1P:** Claiming that we are done if we can make all the numbers odd and then use $n = 2$
- (B) **3P:** Constructing an n that keeps all the odd numbers odd
 - (B1) **1P:** The idea of constructing n using the product of all the numbers, their odd parts or equivalent
- (C) **1P:** Constructing an infinite family of numbers that send a specific even number to an odd number.

A complete solution is awarded 7 points. An incomplete solution is awarded up to 5 points.

8. There are 2026 white balls and 2026 black balls arranged in a row in some order. A *balanced segment* is a non-empty segment of consecutive balls that contains the same number of white balls as black balls. A *balanced removal* is the operation of selecting a balanced segment S , removing all the balls in S , and shifting left the balls that were to the right of S so that the remaining balls again form a contiguous row.

Determine the smallest positive integer N satisfying the following property: for every starting configuration of balls and for every integer k satisfying $0 \leq k \leq 2026$, there exists a sequence of at most N balanced removals after which there are exactly $2k$ remaining balls.

Solution

Answer: $N = 2$

Upper bound: Consider the function $f(x)$, where $f(a)$ is the number of white balls minus the number of black balls in the first a balls. Notice that $f(0) = f(4052) = 0$. We will give a process for constructing any sized sequences of length $2k$ using at most two balanced removals.

Lemma 1: If $f(a) = f(b) = 0$ and for all $a < i < b$ we have that $f(i) > 0$ then it is possible to remove a sequence of any even length between 0 and $b - a$ in one operation.

Assume we want to remove a sequence of even length $2k$. Consider $f(a + i)$ and $f(a + 2k + i)$. If for some $0 \leq i \leq b - a - 2k$ we have that $f(a + i) = f(a + i + 2k)$ then removing the sequence of balls from $a + i$ to $a + i + 2k$ is a balanced removal. Plot the function f and notice that $(a + 2k, 0) = (a + 2k, f(a))$ is in the area created by the function f and the x-axis. Now observe $(b + 2k, f(b))$ is outside said area. Since f is continuous by discrete continuity there exists an i such that $(a + i + 2k, f(a + i))$ is on the boundary of the area. Since $f(a + i) > 0$ it must be on the boundary created by f . Therefore $f(a + i) = f(a + i + 2k)$ for some i and we can perform a balanced removal. Lemma 1 proven.

Notice that Lemma one can also be applied if all the values between $f(a) = 0$ and $f(b) = 0$ are negative.

Now consider the sequence of roots of f to be $0 = a_1 < a_2 < \dots < a_m = 4050$. To get a sequence with exactly $2k$ balls left choose the maximum i such that $f(a_i) = 0$ and $a_i \leq 2k$. Let the first balanced removal be the segment from 0 to a_i . This is balanced as $f(0) = f(a_i) = 0$. Now notice that $a_{i+1} < 2k$ and therefore we can apply lemma one on the sequence from a_i to a_{i+1} to remove exactly $4052 - 2k - a_i$ since it is less than $a_{i+1} - a_i$ as all values between $f(a_i)$ and $f(a_{i+1})$ are all positive or all negative. Therefore we have removed exactly $a_i + (4052 - 2k - a_i) = 4052 - 2k$ balls and we have exactly $2k$ balls left in two balanced removals.

Lower bound: To show that one operation is not always enough, consider the arrangement of 2026 white balls together in the middle and 1013 black balls on each side of the central group of white balls and assume we have $k = 1$ and want to have exactly 2 balls left after one operation. Notice that if we want to remove more than 1013 black balls in one operation then we must remove all 2026 central white balls as there are not enough black balls on each side. Therefore in one balanced removal it is not possible to have 2 balls left as we either remove all the balls or at most 1013 white balls.

Therefore $N = 2$ is the solution.

Marking Scheme

- (A) **1P:** Proving that $N > 1$.
- (B) **4P:** Proving Lemma 1 or an equivalent statement
 - (B1) **1P:** Stating the lemma or an equivalent statement in a clear way.
 - (B2) **1P:** Introducing the function $f(x)$ which is the difference of white and black balls in the first x balls.
- (C) **1P:** Finishing the problem assuming Lemma 1 is proven.